A verified infinite obstruction to a prime plus two powers of two

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Result and interpretation. With the conventional prime-number meaning of p, the printed conjecture is false: for every $L \in \mathbb{N}$ there is an odd $N > L$ such that

$$N \neq p + 2^a + 2^b \qquad \text{for every prime } p \text{ and every } a, b \in \mathbb{N}.$$

Thus zero exponents and repeated exponents are included. If p meant an arbitrary integer, the statement would instead be trivial; that is not the prime-sum interpretation treated here. The construction adapts Crocker's classical proof [1]. Every assertion below has an ordinary Lean proof; finite Python calculations were used only to find certificates.

The explicit arithmetic data. The last page lists 28 triples (d_i, m_i, q_i). Put

$$Q = 8191 \prod_{i=1}^{28} q_i, \qquad D = 16Q, \qquad B_r = \begin{cases} 45592577 & r = 10, \\ 2^{2^r} + 1 & r \neq 10, \end{cases} \qquad G = \frac{2^{1024} + 1}{45592577}.$$

The following identities and inequalities are verified over the natural numbers:

$$1 < q_i, \quad q_i \mid D, \quad m_i \mid 720, \quad 2^{m_i} \equiv 1 \pmod{q_i}, \qquad D < G, \quad G \equiv 1 \pmod{16},$$
$$2^{9360} \equiv 1 \pmod{Q}, \qquad \gcd(2^{4096} - 1, Q) = 1,$$
$$\forall u \in \{0, \ldots, 719\}\ \exists i \quad u \equiv d_i \pmod{m_i}, \qquad q_i + 2^j \not\equiv 0 \pmod{8191} \quad (0 \le j < 13).$$

No primality of the q_i is assumed or needed. The concrete integer C is specified by concatenating the following decimal lines:

```
14105361477916114570737258104493241655354019 3
44940573425140720525837701276167724450285261 3
25054855422208435392507600721311
```

It satisfies $0 \le C < D$ and

$$C \equiv 15 \pmod{16}, \qquad C \equiv 0 \pmod{8191}, \qquad C \equiv 2^{d_i} \pmod{q_i}.$$

An infinite sequence of actual integers. For $k \ge 0$ take $n = 12(k+1)$ and $A_n = \prod_{r<n} B_r$. The Fermat product identity and the one replaced factor give

$$A_n G = 2^{2^n} - 1, \qquad A_n \ge 2^n, \qquad A_n \equiv 15 \pmod{16}.$$

Since $2^n \equiv 4096 \pmod{9360}$, the displayed modular certificates imply $\gcd(2^{2^n} - 1, Q) = 1$. Oddness also gives coprimality with 16; hence $\gcd(A_n, D) = 1$. Define w_k as the least nonnegative residue of $A_n^{-1} C$ modulo D, and set $N_k = w_k A_n$. Then

$$0 < w_k < D < G, \quad 2^n \le N_k < 2^{2^n} - 1, \quad N_k \equiv C \pmod{D}.$$

Positivity follows from $C \equiv 15 \pmod{16}$, so $w_k \neq 0$. In particular, N_k is odd and $N_k > k$.

Excluding one power. Suppose $N_k = p + 2^a$. The covering table supplies i with $a \equiv d_i \pmod{m_i}$. Thus $q_i \mid p$. Because $q_i > 1$ and p is prime, $p = q_i$, even when the chosen modulus q_i is not prime. Reduction modulo 8191 would now give $q_i + 2^{a \bmod 13} \equiv 0$, contradicting the certificate for $0 \le j < 13$. This also excludes two equal powers, since $2^a + 2^a = 2^{a+1}$.

Excluding distinct positive powers. Suppose $N_k = p + 2^a + 2^b$ with $a > b > 0$. The upper bound on N_k implies $a < 2^n$. Write $a - b = 2^r m$ with m odd. Then $r < n$, and

$$2^{2^r} + 1 \mid 2^{a-b} + 1, \qquad B_r \mid N_k, \qquad B_r \mid 2^b(2^{a-b} + 1).$$

Consequently $B_r \mid p$ and $B_r = p$. The product A_n contains both B_r and $B_2 = 17$ if $r \neq 2$; if $r = 2$, it contains $B_0 B_2 = 51$. These facts, together with $B_r > 1$, give $B_r + 16 < A_n \le N_k$. Therefore $2^a + 2^b > 16$, which forces $a \ge 4$.

But B_r is congruent to 3, 5, or 1 modulo 16, respectively as $r = 0$, $r = 1$, or $r \ge 2$. Also $2^a \equiv 0 \pmod{16}$ and $2^b \equiv 0, 2, 4$, or $8 \pmod{16}$. No sum of these residues is 15. This contradicts $N_k \equiv 15 \pmod{16}$. Interchanging a, b covers the other order.

Zero exponents. If exactly one exponent is zero, say $a = 0 < b$, then the odd number $N_k = p + 1 + 2^b$ forces the prime p to be 2. This would yield $2^b \equiv 12 \pmod{16}$, again impossible. Both zero exponents were already excluded by the one-power argument with exponent 1. The exclusions therefore hold for every pair of nonnegative exponents.

The quantified conclusion. For an arbitrary cutoff L, use $k = L$. The inequalities $L < n < 2^n \le N_k$ give a larger odd integer with no permitted representation. This negates the printed "every sufficiently large" assertion, rather than only checking a single large example.

Formal objects and verification. `Main.lean` defines the actual integer data, products, modular inverse, and sequence N_k. The prime predicate is Mathlib's `Nat.Prime`; the modular inverse is the genuine inverse in `ZMod D`. The covering calculation uses ordinary kernel-checked `decide`; large integer identities use `norm_num`. No `native_decide`, extra axiom, or admitted proof is used. The final declarations state the arbitrary-cutoff obstruction and the negation of eventual representability.

The companion project pins Lean 4.33.0 and Mathlib `db584cd6d46c92f209a44c0f1c829460d327499d`. Its evidence records a new build, an all-namespace ordinary axiom audit, and the official fresh-kernel replay. The final submitted package records their actual outcomes; production timings are not controlled efficiency measurements.

References

[1] R. Crocker, *On the sum of a prime and of two powers of two*, Pacific Journal of Mathematics **36** (1971), no. 1, 103–107. doi:10.2140/pjm.1971.36.103. The original five-page argument was read; its printed pp. 104–107 were also rendered and inspected.

The finite covering certificate

Here d_i is an exponent residue, m_i its period, and q_i a divisor modulus. All 720 residues are covered, and all listed modular identities are checked in Lean. The last modulus is composite; none of the proof requires these moduli to be prime.

i	d_i	m_i	q_i
1	0	3	7
2	0	5	31
3	1	9	73
4	1	10	11
5	8	12	13
6	8	15	151
7	4	18	19
8	7	20	41
9	5	24	241
10	29	30	331
11	2	36	37
12	14	36	109
13	17	40	61681
14	34	45	631
15	43	45	23311
16	13	48	97
17	37	48	673
18	16	60	61
19	19	60	1321
20	26	72	433
21	62	72	38737
22	52	90	18837001
23	37	120	4562284561
24	49	144	577
25	121	144	487824887233
26	103	180	181
27	106	180	54001
28	229	360	792475053262249686120362188 81

The retained factor 45592577 divides $2^{1024}+1$ and is 1 modulo 16. Every other B_r is the full Fermat number $2^{2^r}+1$. Only the explicit modular identities, divisibility and bounds stated in the proof are used; the construction does not invoke an unproved factorization or a general primitive-prime-divisor theorem.

The proof is a formalization of a classical counterexample method, with the finite certificates and the zero-exponent cases made explicit. It makes no novelty claim for the underlying number-theoretic result. Organizer acceptance remains pending.